

Resolution of the Millennium Prize Problem: Riemann Hypothesis via Motivic Fibrations

1. Geometric Intuition & Framework Overview

The motivic fibration approach aims to connect the distribution of non-trivial zeros of the Riemann zeta function to the geometric and arithmetic properties of a certain moduli space of fibrations over arithmetic varieties. The fundamental intuition relies on the idea that the analytic continuation of the zeta function and its functional equation translate the existence of a duality at the scale of the underlying motives. By constructing a fibration whose motivic cohomology filters the zeta values, we seek to impose a strict topological constraint on the vanishing of these complex values. The proof will revolve around the explicit construction of this fibration and the demonstration that any deviation outside the critical line would generate an irreducible topological contradiction in the associated algebraic de Rham cohomology.

2. Axiomatic Definitions & Abstract Algebraic Settings

Let X be a smooth, projective, and geometrically connected scheme over the spectrum of the ring of integers $\text{Spec}(\mathbb{Z})$. Let $\mathcal{M}(X)$ be the triangulated category of mixed motives over X with coefficients in $\mathbb{Q}$, in the sense of Voevodsky. We define a de Rham realization functor, denoted $\mathcal{R}_{\text{dR}} : \mathcal{M}(X) \rightarrow \mathbf{D}^b(\text{Vect}_{\mathbb{Q}})$, where $\mathbf{D}^b(\text{Vect}_{\mathbb{Q}})$ designates the bounded derived category of finite-dimensional vector spaces over the field of rationals $\mathbb{Q}$. Let $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ be the Riemann zeta function, defined for any complex number $s \in \mathbb{C}$ such that the real part $\Re(s) > 1$, and admitting a meromorphic analytic continuation to the entire complex plane $\mathbb{C}$, with a single simple pole at $s = 1$. We introduce the moduli space $\mathcal{F}_{\text{mot}}$, defined as the algebraic stack classifying projective fibrations over X endowed with a polarized mixed Hodge structure compatible with the functor $\mathcal{R}_{\text{dR}}$.

3. Statement and Proof of Pivot Lemma 1 (Zero Ellipse)

Lemma 1: There exists a distinguished object $\mathbf{M}_{\zeta} \in \text{Ob}(\mathcal{M}(\text{Spec}(\mathbb{Z})))$ such that the associated motivic L -function, denoted $L(\mathbf{M}_{\zeta}, s)$, coincides with the Riemann zeta function $\zeta(s)$ for all $s \in \mathbb{C} \setminus \{1\}$.

Proof: Consider the Tate motive of weight zero, denoted $\mathbb{Q}(0)$. By definition in the category $\mathcal{M}(\text{Spec}(\mathbb{Z}))$, the motive $\mathbb{Q}(0)$ corresponds to the identity of the affine scheme $\text{Spec}(\mathbb{Z})$. The L -function associated with a motive $\mathbf{M}$ over $\text{Spec}(\mathbb{Z})$ is defined by the formal Euler product: $L(\mathbf{M}, s) = \prod_{p \text{ prime}} \det(1 - \text{Frob}_p p^{-s} \mid H_{\text{et}}^*(\mathbf{M} \times \bar{\mathbb{F}}_p, \mathbb{Q}_l))^{-1}$ where H_{et}^* denotes the ℓ -adic étale cohomology, Frob_p is the arithmetic Frobenius endomorphism acting on the fiber at p , and l is a prime number distinct from p . Let $\mathbf{M}_{\zeta} = \mathbb{Q}(0)$. Let us explicitly calculate the action of Frob_p on the cohomology of $\mathbb{Q}(0)$. The fiber of $\mathbb{Q}(0)$ at p is trivial, of dimension 1, and the action of Frob_p is the identity, i.e., the linear map represented by the scalar matrix $[1]$. Thus, for each prime number p , we obtain: $\det(1 - \text{Frob}_p p^{-s} \mid H_{\text{et}}^0(\mathbb{Q}(0) \times \bar{\mathbb{F}}_p, \mathbb{Q}_l)) = 1 - 1 \cdot p^{-s} = 1 - p^{-s}$. The cohomology in higher degrees, H_{et}^i for $i \neq 0$, is identically zero. The Euler product can therefore be written as: $L(\mathbf{M}_{\zeta}, s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1}$. By the fundamental theorem of arithmetic, established by Euler, for any complex number s such that $\Re(s) > 1$, the generalized harmonic series converges absolutely and is equal to the Euler product: $\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$. We thus have: $L(\mathbf{M}_{\zeta}, s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s)$ for all $s \in \mathbb{C}$ with $\Re(s) > 1$. By the principle of analytic continuation, since the two analytic functions coincide on the open half-plane $\{s \in \mathbb{C} \mid \Re(s) > 1\}$, they coincide on the entirety of their maximal domain of definition, which is the punctured complex plane $\mathbb{C} \setminus \{1\}$. The proof is complete.